

DiaTool-DPO: Multi-Turn DPO for Tool-Augmented LLMs

Enhancing dialogue flow via Direct Preference Optimization with automatic paired-trajectory construction

A novel DPO-based method that models TA-LLM interactions as an MDP with 5 dialogue states and 3 query types, automatically constructing preference pairs to improve slot-filling and tool call rejection without human labeling.

SFT Alone Fails to Teach Slot-Filling and Tool Rejection

Problem Statement

TA-LLMs must decide whether to ask follow-up questions, make a tool call, or reject inappropriate requests.

SFT with expert trajectories alone cannot learn robust conversational control.

Missing arguments and out-of-scope queries lead to hallucinated tool invocations.

Three Failure Modes of SFT-Only Training

1. Failing to ask follow-up questions for missing arguments → incomplete slot-filling.
2. Hallucinated tool invocations when arguments are underspecified.
3. Failing to reject out-of-scope or unavailable tool calls.

Insight: Preference Optimization (DPO) can teach the model to prefer correct dialogue flows over incorrect ones by constructing paired trajectories of chosen vs. rejected behavior.

Modeling TA-LLM Dialogue as a 5-State Markov Decision Process

1. Initial State

No history. Out-of-scope requests return rejection and remain here.

2. Tool Selected w/o Complete Slots

Tool identified but required parameters missing. Slot-filling QA occurs.

3. Tool Selected w/ Complete Slots

Tool selection and all argument values determined.

4. Wait for Tool Response

Tool call executed, awaiting results.

5. Complete

Tool results received; completion message generated for user.

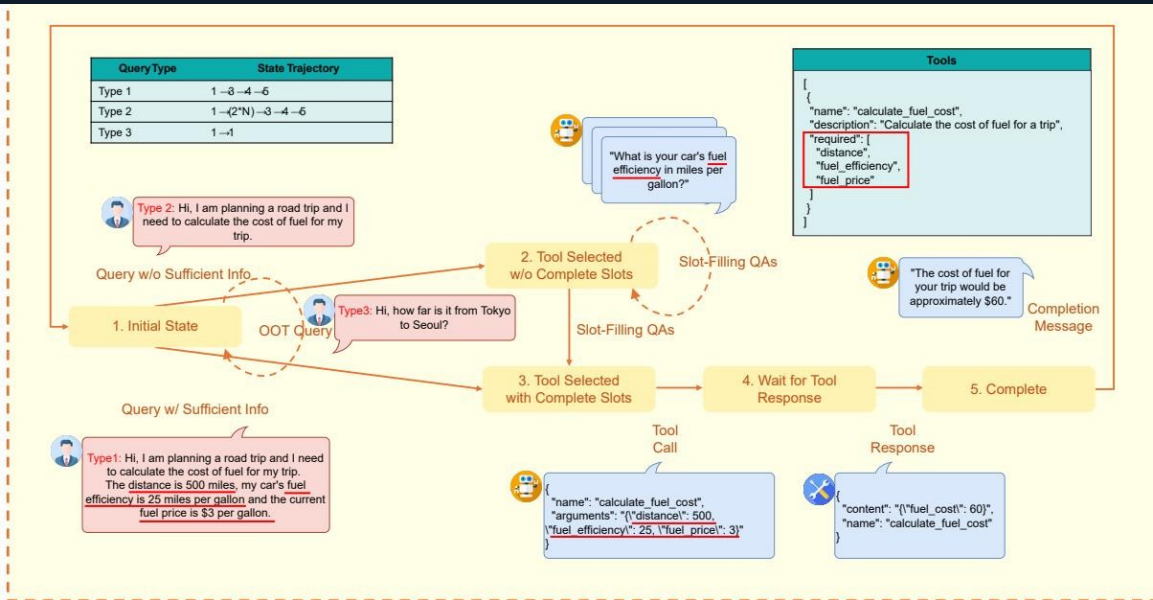


Figure: Visualization of five internal states and state trajectories for three query types

Three Query Types Define Distinct State Trajectories

Query Type	State Trajectory	Description	Example
Type 1	$1 \rightarrow 3 \rightarrow 4 \rightarrow 5$	Query contains all required arguments; no slot-filling needed.	"Book a flight from NYC to LAX on May 10th" (all slots filled).
Type 2	$1 \rightarrow (2 \times N) \rightarrow 3 \rightarrow 4 \rightarrow 5$	Query lacks some arguments; slot-filling QA repeats N times.	"Book a flight to LAX" $\rightarrow$ system asks for origin and date.
Type 3	$1 \rightarrow 1$	Requested functionality unavailable; tool call must be rejected.	"Send a fax" when no fax tool exists in the API.

Key Insight: By classifying every user query into one of three trajectories, DiaTool-DPO can systematically generate chosen (correct) and rejected (incorrect) trajectory pairs for DPO training.

Automatic Paired Trajectory Construction Without Human Labels

Each pair teaches a specific dialogue-control lesson. Dataset composition on FunctionChat-Bench:

Query	Chosen Trajectory	Rejected Trajectory	Learning Lesson	Count
Type 1	1→3→4→5	1→2→3→4→5	Prevent redundant slot-filling	2,089
Type 1	1→3→4→5	1→(2×N)→3→4→5	Prevent redundant slot-filling	562
Type 1	1→3→4→5	1→(2×M)→3→4→5	Prevent redundant slot-filling	2,530
Type 1	1→3→4→5	1→1	Tool call accept	2,090 / 562
Type 2	1→2→3→4→5	1→3→4→5	Prevent slot hallucination	2,089
Type 2	1→(2×N)→3→4→5	1→3→4→5	Prevent slot hallucination	562
Type 2	1→(2×N)→3→4→5	1→(2×M)→3→4→5	Prevent slot hallucination	2,530
Type 2	—	1→1	Tool call accept	2,089 / 562
Type 3	1→1	1→3→4	Tool call reject	567
Type 3	1→1	1→(2×N)→3→4	Tool call reject	562

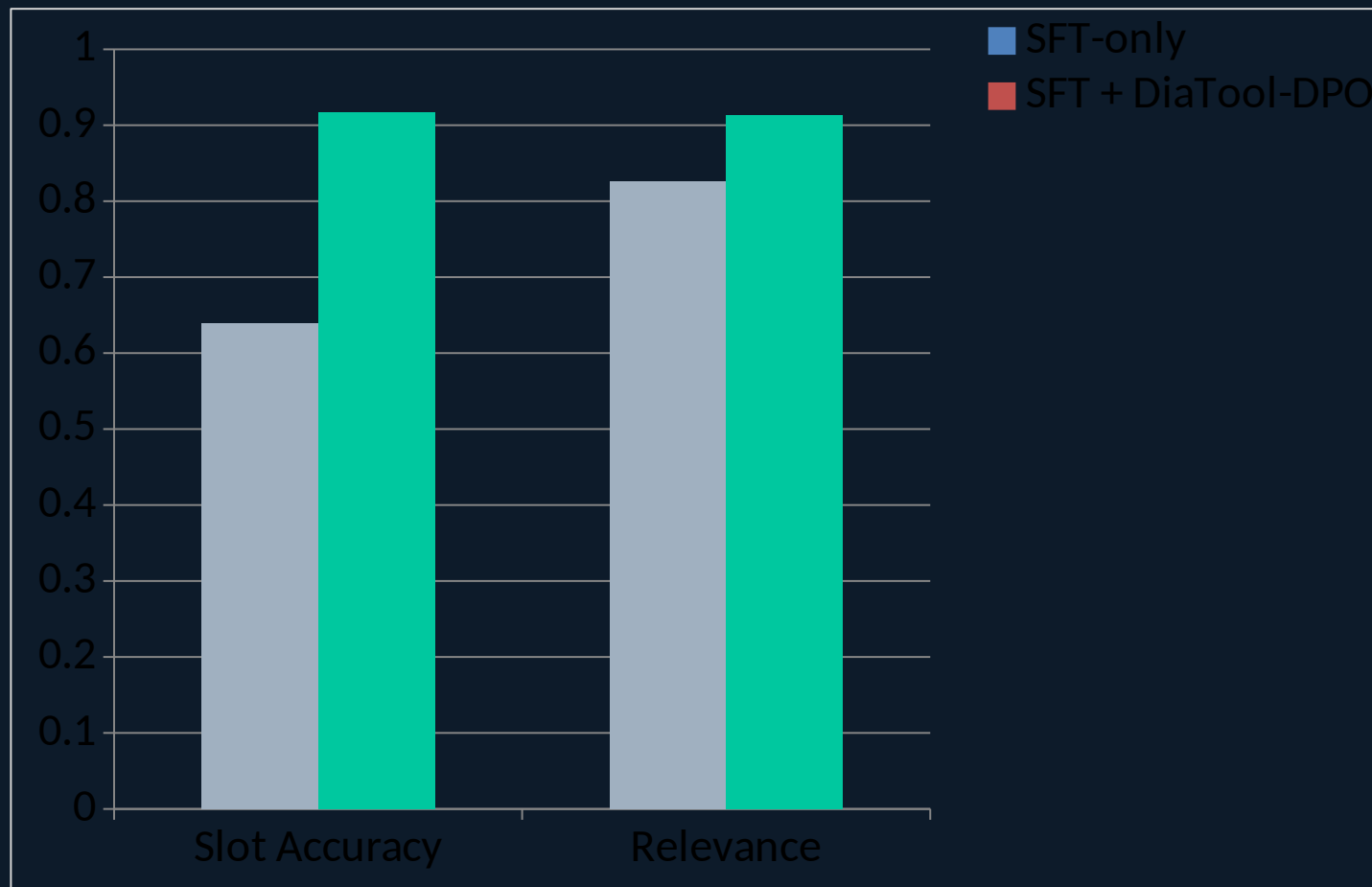
SFT + DiaTool-DPO Achieves Best Macro Average Across All Models

Results on FunctionChat-Bench. Bold = best per model.

Model	Method	Call	Completion	Slot	Relevance	Micro Avg	Macro Avg
Prop.-8B	DiaTool-DPO-only	0.314	0.700	0.833	0.609	0.575	0.614
Prop.-8B	SFT-only	0.900	0.916	0.694	0.913	0.870	0.856
Prop.-8B	SFT + DiaTool-DPO	0.886	0.929	0.833	0.826	0.884	0.868
Prop.-3.1B	DiaTool-DPO-only	0.357	0.551	0.528	0.391	0.455	0.457
Prop.-3.1B	SFT-only	0.771	0.817	0.750	0.826	0.790	0.791
Prop.-3.1B	SFT + DiaTool-DPO	0.743	0.871	0.833	0.826	0.765	0.818
LLaMA-3-8B	DiaTool-DPO-only	0.029	0.449	0.056	0.261	0.205	0.199
LLaMA-3-8B	SFT-only	0.843	0.957	0.639	0.826	0.844	0.816
LLaMA-3-8B	SFT + DiaTool-DPO	0.857	0.929	0.917	0.913	0.905	0.904

LLaMA-3-8B: +43.5% Slot Accuracy and +10.5% Relevance with DiaTool-DPO

Adding DiaTool-DPO to SFT yields the largest gains on slot-filling and tool-rejection metrics.



Slot Accuracy
0.639 → 0.917
(+43.5%)

Relevance
0.826 → 0.913
(+10.5%)

Near GPT-4o performance: 94.8% information gathering • 91% tool call rejection

Key Takeaways: RL-Based Dialogue Control for Production TA-LLMs

MDP formulation enables systematic preference pair construction

Defining 5 internal states and 3 query types allows automatic generation of chosen/rejected trajectory pairs without human annotation.

Automatic dataset construction eliminates human labeling

DiaTool-DPO constructs ~10K paired trajectories purely from state-transition rules, scaling to new domains at zero annotation cost.

DPO complements SFT — it does not replace it

DiaTool-DPO-only performs poorly across all models; the synergy of SFT + DPO yields the best balanced (Macro Avg) results.

Largest gains in slot-filling and tool rejection

LLaMA-3-8B improves Slot by +43.5% and Relevance by +10.5%, reaching near-GPT-4o conversational tool-use performance.

Language-agnostic and production-ready

Primary experiments in Korean generalize to English, opening broad applicability for multilingual production TA-LLMs.